

Admissible Hubs and Newton-Flow Ray Separation

Checked dynamical inputs and the remaining metric problem for Erdős #1041

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ABSTRACT

Let $f(z) = \prod_i (z - z_i)$ be monic with roots in the closed unit disc. Erdős, Herzog, and Piranian ask whether two roots can always be joined by a curve of length less than 2 inside $\{|f| < 1\}$. The global assertion remains open.

Two exact inputs isolate the obstruction. At every critical point, two nearest-root occurrences have total Euclidean distance at most 2; Lean checks the reciprocal critical balance and disk inverse-square estimate used by the ordinary polynomial-level argument. This does not prove containment. An exact quintic has a nearest straight spoke that leaves $\{|f| < 1\}$, and an exact cubic has an escaping midpoint on every straight root-pair segment. Along the complex Newton field $-f/f'$, the value satisfies $f(z(t)) = e^{-t}f(z(0))$. Lean checks this value equation, the resulting positive-ray obstruction to finite saddle connections, and the finite translation and root-retention interfaces.

The remaining metric problem is therefore one of admissible-hub selection. On the ray-separated dense class, one must find a critical hub c whose two inverse-ray arms stay in the relevant component and satisfy

$$\min_{c \text{ admissible}} L(c) \leq 2.$$

At degree five, a critical value at most $1/M_5$, where

$$M_5 = (1 - t_*)(1 + t_*)^3 \sqrt{16t_*^2 - 4t_* + 1}, \quad t_* = \frac{5}{16} + \frac{3\sqrt{105}}{80}, \quad 1/M_5 = 0.2760461 \dots,$$

would give two contained nearest-root spokes of total length at most 2. The near-Fekete reduction removes radial deficits from this containment question, but neither the general hub selection nor its degree-five form is proved. The threshold and no-go witnesses are ordinary proofs, exact certificates, or computations, not additional kernel-checked theorems. Coefficient genericity, component stability, planar topology, and metric gluing remain open.

Contribution. The paper checks exponential Newton-value decay, parameterises positive-ray collisions, constructs arbitrarily small translations avoiding all collisions in a finite family, and proves root retention under a quantified constant perturbation. It also records the current near-Fekete reduction and the degree-five residual.

**Authorship and AI use.* Will Cook built and directed the research infrastructure and reviewed the claims when he could. AI agents did most of the research and drafting. Cook did not independently verify every claim.

Relation to the open problem. These are dynamical, perturbative, and frontier inputs. They do not repair the global topology and metric gluing needed for a curve of length less than 2, so Problem #1041 remains open. The degree-five target and the no-go witnesses are boundaries around the open theorem, not a hidden closure.

Executable review object. Comparator selects the finite-family small-translation theorem and the quantified root-retention theorem. The frontier section is selected as mathematical orientation; its ordinary proofs, exact certificates, and computations remain separate evidence classes. The repository’s external-verification job compares these exact Lean propositions with separately declared challenge statements and an axiom budget, then asks Lean’s kernel to check the submitted proofs. The [formalisation manifest](#) and the commit-bound CI receipt record that check; they do not assess novelty, significance, or whether the original problem is solved.

The two selected perturbation interfaces have the following exact boundaries. For a finite type ι , an injective family of critical values $c: \iota \rightarrow \mathbb{C}$, and every $\varepsilon > 0$, the row `exists_small_translation_separating_arguments` supplies a shift of norm less than ε such that every translated value is nonzero and every two distinct translated values lie on different positive rays. Separately, for a monic split polynomial of positive degree whose roots have norm at most ρ , the row `constant_perturbation_roots_in_unitDisk` requires $\rho \geq 0$, $\varepsilon > 0$, the margin

$$((\text{natDegree } f + 1)\varepsilon)^{1/\text{natDegree } f} + \rho < 1,$$

and a shift of norm less than ε ; it then places every root of $f + C(\text{shift})$ in the open unit disc. These are finite perturbation and stability interfaces only: neither row supplies the missing global topology or the length-2 gluing argument.

1 The problem

Problem 1.1 (Erdős #1041). Let $f(z) = \prod_{i=1}^n (z - z_i)$ be monic with $z_i \in \mathbb{D}$, the open unit disc. Show that two of the roots can be joined by a curve of length less than 2 lying in the open lemniscate $E = \{z \in \mathbb{C} : |f(z)| < 1\}$.

Numbering and current status follow Bloom’s Erdős problem catalogue [1]. The problem is open. The original source is Problem 5 on printed p. 139 of Erdős–Herzog–Piranian [2, p. 139]; the preceding paragraph records the known input that one component of the lemniscate contains at least two zeros.

Recent work on polynomial lemniscates separates component counts from metric path questions. Ghosh and Ramachandran characterize the number of components through critical points and critical values [3, Lemma 7]; for the binomial family $z^n - a$, the condition $|a| < 1$ puts the filled unit lemniscate in the connected regime. Connectedness alone gives no path-length bound. Our dynamical terminology is also standard: Sutherland calls $\dot{z} = -f(z)/f'(z)$ the continuous Newton flow and observes that $f(z(t))$ moves on a straight radial line [4, p. 42]. We use this value-space identity, not a claim that the global trajectory graph is a tree.

Two recent manuscripts are relevant. The 48-page manuscript posted by shtuka on 24 March 2026 [5, Theorem 1, p. 1] claims the unrestricted statement. Its Proposition 12

(p. 16, with proof continuing through p. 30) supplies the spanning-tree decomposition used in the final proof. The defect was located publicly in the problem’s discussion thread: on 25 March 2026 Tao observed that the invocation of Lemma 8 there is unjustified and that the flow lines need not organise into connected trees, and on 26 March 2026 the manuscript’s author agreed that the statement of Proposition 12 itself, not only its printed proof, is incorrect, and set the strategy aside. Section 5 records an independent diagnosis of the same failure through the local three-ended saddle model, together with possible repairs. No counterexample to the proposition is exhibited there. Pendyala’s independent June 2026 preprint [6, Thm. 1, p. 1] proves the degree-four case. The quartic theorem does not close the problem.

The object we work with is not the lemniscate directly but the flow that foliates it.

Status. The problem treated here is open, and this note does not close it. Every statement below marked as checked is a proposition that the pinned Lean kernel accepts from the sources this note links to, with no sorry, no added axiom, and no unchecked evaluation. That is a claim about the formal statement, not about its mathematical interest, its novelty, or the original problem. The unresolved obligations are named exactly, in their own section, and none of the finite computations, reductions, or no-go results here removes one of them.

2 The current frontier: what survives near Fekete configurations

The August 29 public research update changes the useful first reading of this problem. It does not add a proof of Erdős #1041; it identifies the geometric regime in which the remaining difficulty is concentrated and kills several attractive but false shortcuts. The source of the update is pinned to commit [f214a6b4](#) and the dated synthesis is [FRONTIER.md](#). The distinctions below are part of the result: a theorem, a refutation, a certificate, and a measurement do not have interchangeable force.

Near-Fekete stability removes the radial distraction

Write $a_i = \rho_i u_i$, with $|u_i| = 1$, and put $D = |\text{disc}(f)|/n^n$. The quantitative Fekete–Hadamard estimate in [NearFeketeRadialAngularSplit.md](#) says that, when $D \geq 1 - \eta$ and $\eta \leq 1/(80n^2)$,

$$1 - \rho_i^2 \leq \frac{n\eta}{n-1}, \quad |a_i - a_j| \geq \frac{2 - 2\sqrt{\eta} - n\eta}{n-1},$$

and the roots lie within $7\sqrt{\eta}$ of a rotated regular n -gon, after a bijection. The mechanism is a normalised Vandermonde Gram determinant: near equality in Hadamard’s inequality forces both the radii and the angular separation to be near extremal. This is an ordinary mathematical result in the public research corpus, not a new Lean declaration.

The associated radial monotonicity theorem is the more important reader conclusion. Under the stronger near-Fekete condition $\eta \leq 1/(10n^4)$ used by this corollary, replacing the radii by one can only increase the values along the corresponding radial spoke. Thus

radial deficits help containment; the unresolved near-Fekete problem is angular. On exact regular-gon directions this becomes unconditional for $n \leq 6$: arbitrary radii in the stated band give

$$|f(s\omega\zeta^i)| \leq 1 - s^n \quad (0 \leq s \leq \rho_i),$$

so every two-radii path is contained. The good spoke in the general on-circle averaging identity may move with s , however; that is why an existence statement at one level does not supply a fixed pair of roots.

A bounded-radius concyclic class

There is nevertheless a clean theorem for a different on-circle mechanism. Let f be monic of degree $n \geq 3$ with distinct zeros on a circle of radius ρ . If $2\rho^n \leq 1$, two adjacent zeros are joined by their straight chord, whose length is at most $2\rho \sin(\pi/n) < 2$, and the chord lies in $\{|f| < 1\}$. (If a zero is repeated, the short-connection conclusion is immediate; the distinct case is the substantive one.)

After normalisation to the unit circle, self-inversive realification and alternation against $z^n - c$ select a zero-free gap on which the polynomial modulus is at most 2; a harmonic normal-derivative argument transfers this arc bound strictly to the chord. The complete ordinary proof is [ConcyclicAlternation.md](#).

This is not a Lean theorem: the [exact-rational checker](#) tests finitely many load-bearing identities and configurations, while the [numerical checker](#) is regression and stress-test evidence. The arc constant 2 is sharp on the regular n -gon, so this chord argument does not reach radii tending to 1; the unrestricted concyclic case and Erdős #1041 remain open.

Cyclic trinomial fibres

There is also a positive structured family in every cyclic lift degree. Fix integers $1 \leq r \leq m$ and $q \geq 1$, and consider the translated polynomial

$$f(z) = (z - h)^{qm} + a(z - h)^{qr} + c.$$

Writing $w = (z - h)^q$ reduces the root equation to $w^m + aw^r + c = 0$. At such a quotient root the middle coefficient can be eliminated exactly: for $0 \leq u \leq 1$,

$$u^m w^m + au^r w^r + c = (1 - u^r)c - (u^r - u^m)w^m.$$

The two coefficients on the right are nonnegative and have total at most one. Lean therefore proves that, if $|w| < 1$ and $|c| < 1$, the complete radial spoke is strictly contained in the open unit lemniscate. This is an all-root statement; no Vieta-small root selection is needed for the containment bound.

For a nontrivial cyclic fibre, two selected displacements y_1, y_2 with $|y_1|, |y_2| < 1$ also satisfy $|y_1| + |y_2| < 2$. The formal source checks the factorization, strict spoke estimate, and this metric budget. It does not yet check the ordinary finite argument selecting a suitable quotient root and two distinct members of its fibre, lifting both quotient spokes, or assembling their union as a path in the relevant component. Thus the Lean theorem is the load-bearing analytic kernel for this translated cyclic-trinomial family, not the complete path theorem and not a proof of unrestricted Erdős #1041.

Coefficient-controlled cyclic tetranomial spokes

The first additional lower monomial admits an exact Abel-tail criterion. Let

$$g(w) = w^m + aw^r + bw^s + c, \quad m > r > s \geq 1,$$

assume all roots of g lie in the open unit disk, and let w_1, w_2 be roots of the two smallest moduli. If

$$|c| + |b||w_2|^s < 1,$$

then the complete radial spokes from w_1 and w_2 to the origin lie in $\{|g| < 1\}$, and their broken line has length strictly below 2. The simpler coefficient condition

$$|b| + |c| \leq 1$$

makes every root spoke safe; the coefficient a is unrestricted.

The mechanism is visible in one identity. At a root w and for $0 \leq u \leq 1$,

$$\begin{aligned} g(uw) &= (1 - u^s)c - (u^s - u^r)(aw^r + w^m) - (u^r - u^m)w^m, \\ aw^r + w^m &= -(c + bw^s). \end{aligned}$$

The three scalar coefficients are nonnegative and sum to $1 - u^m$. Thus the root-dependent budget $|c| + |b||w|^s < 1$, together with $|w| < 1$ and $|c| < 1$, puts every vector in the Abel decomposition strictly inside the unit ball. Lean checks this factorization and the resulting strict spoke theorem under the exact weak exponent hypotheses $1 \leq s \leq r \leq m$; it also checks the coefficient-only corollary above.

There is a stronger, genuinely two-index formal theorem. Let S index a finite family of roots, $N = |S| \geq 2$, and

$$M = \sum_{i \in S} w_i^s.$$

Lean proves the exact signed energy identity

$$\sum_{i \in S} |c + bw_i^s|^2 = N|c|^2 + |b|^2 \sum_{i \in S} |w_i^s|^2 + 2 \operatorname{Re}(\bar{c} b M).$$

Consequently, if every $|w_i| < 1$, $|c| < 1$, and

$$N(|b|^2 + |c|^2) + 2 \operatorname{Re}(\bar{c} b M) < N - 1,$$

then two distinct indices $i, j \in S$ satisfy $|c + bw_i^s| < 1$ and $|c + bw_j^s| < 1$. Feeding those inequalities into the Abel decomposition proves, in Lean, that both complete root spokes lie strictly in $\{|g| < 1\}$. The signed cross term is essential: this interface can certify configurations outside the coefficient-only triangle $|b| + |c| \leq 1$. The formal hypotheses do not require $i \mapsto w_i$ to be injective, so distinct indices need not denote distinct root values; the ordinary two-root consequence requires a root enumeration without repetition.

For $q \geq 2$ and $f(z) = g((z - h)^q)$, the ordinary regular-fibre mean-square identity puts every quotient root inside the unit disk. Lifting two selected quotient spokes gives a path through h whose two displacement lengths sum to less than 2. The new Lean theorem assumes the finite root family, its signed moment M , and the displayed budget; it does not derive that budget from arbitrary tetranomial coefficients, prove that the supplied family is a complete root multiset, perform cyclic-fibre lifting, or construct the final path object. Without the signed-moment, root-dependent, or coefficient-only budget, the tetranomial case remains open; this family does not solve unrestricted Erdős #1041.

Translated quartic quotient fibres

Pendyala's degree-four theorem can also be lifted through every nontrivial cyclic power. Let P be a monic quartic, let $q \geq 2$, and set

$$f(z) = P((z - h)^q).$$

If all listed zeros of f lie in the open unit disk, averaging the squared moduli over each complete q -point fibre shows that every quotient root w of P satisfies $|w| < 1$. Pendyala's chord-or-radial proof then supplies the quotient geometry. The extra issue is metric: a short chord in the w -plane need not lift isometrically through $y \mapsto y^q$.

Put $\alpha = 1/q$. Along a quotient chord whose supporting line has distance d from the origin, the root-lift density is controlled pointwise by

$$(\sqrt{d^2 + x^2})^{\alpha-1} \leq x^{\alpha-1} \quad (x > 0),$$

and Lean checks the exact primitive

$$\alpha \int_0^A x^{\alpha-1} dx = A^\alpha.$$

Splitting at the perpendicular foot, and at the origin when the chord crosses it, yields the ordinary power-map chord-lift estimate

$$\text{length}(\widetilde{[a, b]}) \leq |a|^{1/q} + |b|^{1/q}.$$

The formal source also checks the strict endpoint consumer: if $0 \leq a, b < 1$, $\alpha > 0$, and a candidate length L is at most $a^\alpha + b^\alpha$, then $L < 2$. Thus both Pendyala branches survive the power-map lift, giving the asserted short path for every translated quartic quotient fibre, in each degree $4q \geq 8$.

The attribution and proof boundary are exact. Pendyala proves the quartic geometric theorem and its four-point radial lemma. The local Lean module checks the antitone density inequality, its integral, the strict powered endpoint budget, and the final length fan-in. It does not formalize Pendyala's geometric lemma, the continuous covering-space construction of the root lift, or the ordinary chord/radial case assembly. This is a structured all-scale family, not a proof of unrestricted Erdős #1041.

The degree-five target is now explicit

The degree-five analysis sharpens the two-nearest-spoke threshold. If a critical point c satisfies $|f(c)| \leq 1/M_n$, then both straight segments to the two nearest roots remain in $\{|f| \leq 1\}$ and their total length is at most 2. The envelope is obtained by maximising the exact product along the second spoke; at degree five,

$$M_5 = (1 - t_*)(1 + t_*)^3 \sqrt{16t_*^2 - 4t_* + 1}, \quad t_* = \frac{5}{16} + \frac{3\sqrt{105}}{80}, \quad \frac{1}{M_5} = 0.2760461 \dots$$

This is an 11-fold improvement over the earlier deep-low threshold, but it does not assert that every quintic has a critical value below this level.

The exact remaining degree-five statement is therefore not “improve the constant” but the following selection problem:

For every monic quintic with roots in the closed unit disc, there is a critical point c with $|f(c)| \leq 1$ and two roots a, b such that $|f| \leq 1$ on $[c, a] \cup [c, b]$ and $|c - a| + |c - b| \leq 2$.

The public file calls this (SPOKE-5). Together with the existing two-segment reduction it would settle degree five, but (SPOKE-5) is not proved. Coverage experiments put the residual in rapid, nearly simultaneous merging near the regular pentagon: the generic sampled families were covered, whereas the surviving band has every named merge threshold active and D near 1. Those percentages locate the work; they are not a theorem.

At the same degree, the terminal connected component admits a sharper area identity. If $K_t = \{|f| \leq t\}$ is connected and the centred exterior map is

$$\psi(\zeta) = t^{1/n}\zeta + a_0 + \sum_{k \geq 1} a_k \zeta^{-k},$$

then Grönwall's identity gives

$$\text{Area}(K_t) = \pi \left(t^{2/n} - \sum_{k \geq 1} k |a_k|^2 \right), \quad a_1 = -\frac{c_{n-2}}{n t^{1/n}} \quad \text{after centring.}$$

It replaces a Pólya upper bound at that terminal node, but extending it to proper components and connecting it to the parent theorem remains open. The degree-five statements and their evidence classes are recorded in [Degree5AssemblyAndSharpenedCuts.md](#).

The no-go boundaries force hub selection

The same frontier is useful because it closes off three misleading readings. First, the strict-argmin first-merge hub need not have two-arm length below 2. A degree-five tie-locus witness, moved into the open unit disc, has

$$L(c_*) = 2.0573432753 \dots > 2,$$

with a 50-digit two-instrument reconstruction and a margin about 97 times that of the earlier degree-four witness. This is computational certificate evidence, not an exact rational theorem; the parent remains viable because a different hub at the same configuration supplies the relevant pair.

Second, the aggregate estimate $\sum_c L(c) \leq 2(n-1)R$ is refuted at degree four on an open violating region and at one degree-five witness. Its algebraic factor

$$\sum_{k=1}^{n-1} |f(c_k)|^{1/n} \leq (n-1)R$$

survives as a separate conjecture, but the metric factor is not automatic. These two boundaries, including the distinction between the surviving algebraic half and the dead aggregate, are in [MinimalHubArmBudgetRefutation.md](#) and [SeparatrixAggregateReduction.md](#).

Third, the origin is not a uniform near-Fekete hub. An exact rational quintic with roots scaled by $999999/1000000$ has four of five origin spokes escaping $\{|f| \leq 1\}$ at explicit

rational sample points, even though $1 - D = 3.19991 \cdot 10^{-4}$. Consequently no fixed η -neighbourhood of the Fekete locus can force two contained origin spokes in degree five. The mechanism is angular: the first several Fourier modes can point against different spokes. The certificate is exact for the displayed witness; the claim that the phenomenon persists throughout a limiting family is computational. The full witness and replay route are in [NearFeketeRadialAngularSplit.md](#).

The surviving carrier and the actual open endpoint

What survives these refutations is not a fixed hub but a selector. On the ray-separated dense class, the public frontier records the canonical open row

$$\min_{c \text{ admissible}} L(c) \leq 2,$$

where admissibility includes the two inverse-ray arms being contained in the relevant lemniscate component. Its measured values remain below 2 at the refuting configurations, and the existing attachment and lower semicontinuity reductions would turn this row into the parent theorem. No proof of the row is currently available. In particular, the nonzero hub that rescues the origin-spoke witnesses is evidence for the selector, not a universal construction.

The honest current endpoint is therefore a containment selector with three separate obligations: choose the hub, keep both arms inside the component, and control the metric fan-in. The Lean module cited elsewhere in this note checks the Newton value equation, ray-collision avoidance, and root-retention inputs; the new frontier files above are ordinary proof, exact-certificate, and computational research records, not formal authority for (SPOKE-5), the no-go generalisations, or the parent theorem. Keeping those authority boundaries visible is part of making the frontier reusable.

3 The Newton value equation

Away from the critical set, define the complex Newton field

$$N(z) = -\frac{f(z)}{f'(z)}.$$

Let $z(t)$ be differentiable with $z'(t) = N(z(t))$, and put $w(t) = f(z(t))$.

Theorem 3.1 (value equation). $w'(t) = -w(t)$.

The computation is one line: $w' = f'(z) z' = f'(z) \cdot (-f(z)/f'(z)) = -f(z) = -w$. The kernel checks it as *newton flow value has deriv at*, together with the differential form of the first integral, *newton flow scaled value has deriv at zero*:

$$\frac{d}{dt}(e^t f(z(t))) = 0, \quad \text{equivalently} \quad f(z(t)) = e^{-t} f(z(0)).$$

Observe what this says about the geometry. The value moves radially inward at exponential rate and never changes argument. The lemniscate $\{|f| < 1\}$ is therefore forward-invariant, and the flow lines are exactly the preimages of rays from the origin.

Corollary 3.2 (ray separation). *The endpoints of any finite Newton-flow connection lie on the same oriented ray from 0. Consequently, if two critical values lie on distinct positive rays, no Newton-flow saddle connection joins the corresponding critical points.*

This is checked in consumer form: the kernel accepts the implication from the hypothesis of distinct rays to the absence of a connection. It is the statement the topology needs, and it is a genuine sharpening of the criterion used in the literature.

4 Arguments, not moduli

It is tempting to arrange a generic perturbation so that the critical values are pairwise distinct, or that their moduli are pairwise distinct, and to conclude that saddle connections are excluded. Neither is enough.

Two distinct critical values can lie on one ray, and two critical values with distinct moduli certainly can: the ray records the argument, and the modulus is exactly the coordinate the flow contracts. By Corollary 3.2 the invariant that excludes connections is the argument. What a perturbation must therefore achieve is pairwise distinct critical-value *arguments*, which is a condition on $n - 1$ points modulo the circle rather than on their positions in the plane.

The cost of that condition is also checked, and it is small.

Theorem 4.1 (ray-collision locus). *Let $a \neq b$ be complex. Every common translation β for which $a + \beta$ and $b + \beta$ lie on the same positive ray has the form*

$$\beta = \frac{ra - b}{1 - r}, \quad r \in \mathbb{R}_{>0}, r \neq 1.$$

So each pair of critical values contributes a one-real-parameter forbidden locus in the translation plane, given in closed form (*translated same positive ray parameterization*). A finite union of such loci has empty interior, which is the shape one wants for an avoidance argument. Turning that into a perturbation of f is not immediate: the translation model must be replaced by an actual perturbation of the roots that keeps them inside $\mathbb{D}$ and preserves the length slack. That is the first open producer of §7.

5 A proof gap in the unrestricted argument

The March manuscript's Proposition 12 claims the following load-bearing statement. For $u = -\log |f|$, a connected component V of $\{u > c\}$ carrying $m \geq 2$ simple zeros, and the stated regularity and Morse hypotheses, it constructs, for every $\varepsilon > 0$, an embedded spanning tree $G_\varepsilon \subset V$ with

$$\text{len}(G_\varepsilon) \leq \frac{1}{2\pi} \int_{2\alpha}^\infty P_V(t) dt + \varepsilon. \quad (5.1)$$

The final theorem uses this estimate, so the issue below cannot be bypassed by calling the proposition auxiliary.

At an interior index-one critical point p , the proof invokes a Morse chart

$$u = \mu + x^2 - y^2$$

and replaces the saddle by a three-ended neighbourhood having one connected lower cross-section and two connected upper cross-sections. That local model is false as written. Because $p \in V$ and V is open, a sufficiently small closed disc around p lies entirely in V . In that disc the full Morse chart has four sectors: two components of $u > \mu$ and two components of $u < \mu$. A global component argument cannot delete one local sector from a disc already contained in V .

This diagnoses a proof step, not the proposition's statement. A repair might cut an adjoining regular annulus along a separatrix or regular flow arc before forming the block, retain a four-pronged saddle neighbourhood and change the assembly, or replace the local construction by the ray-cut decomposition proposed below. Any repair must prove that its connector cost can be made arbitrarily small uniformly in the attachment points and that the repaired blocks still assemble to an embedded tree satisfying (5.1). The shorter descriptions of the same three-ended block do not repair the four-sector topology.

Corollary 3.2 supplies one independent input for a different route: distinct critical-value arguments exclude saddle-to-saddle Newton connections. It does not itself prove the compact planar decomposition, classify all orbit endpoints or provide the metric gluing estimate. Those are separate problems below.

6 Finite evidence

A search was run over random monic polynomials with roots in the unit disc. For each sample the region $\{|f| < 1\}$ was rasterised and shortest grid paths were computed between every pair of roots. The best upper bounds obtained were

degree 5,	500 trials :	1.1052648928
degree 6,	1500 trials :	0.8450414343
degree 8,	1500 trials :	0.6203916714
degree 10,	1500 trials :	0.4303640486.

No counterexample candidate was found, and the measured values sit well below the threshold 2. These are grid distances for the sampled configurations: upper bounds on those samples, not bounds over the family. The apparent decrease with degree is a property of the sample, and we draw no conjecture from it. Future searches should target named failure modes—near-degenerate saddles, thin necks, boundary-critical configurations and almost-connected separatrices—and report those diagnostics. Raster paths remain candidate finders, never continuous certificates.

7 Complements and further questions

The dependency chain has five separate gates. A proof, a minimally corrected hypothesis set, or an explicit polynomial or planar counterexample is a decisive answer to any one of them.

1. Repair or refute the saddle block

Problem 7.1 (corrected local saddle assembly). Under the hypotheses of Proposition 12, replace the invalid three-ended local model by a four-pronged block or by a block formed after a specified annular cut. Prove that for every $\eta > 0$ its connector has total Euclidean length below η , uniformly over the selected attachment points, and that the corrected blocks assemble to an embedded tree satisfying

$$\text{len}(G_\varepsilon) \leq \frac{1}{2\pi} \int_{2\alpha}^{\infty} P_V(t) dt + \varepsilon;$$

or give a polynomial or harmonic planar counterexample to that statement.

The full-disc model $x^2 - y^2$, an annulus in which lower branches rejoin, two saddles joined by a separatrix and simultaneous saddle levels are mandatory tests. Repeating the one-lower/two-upper assertion does not answer the problem.

2. The compact ray-cut decomposition

Let p have simple roots, simple nonzero critical points and let $u = -\log |p|$. For regular values $c < T$, take

$$M_{c,T} = \overline{V \cap \{c < u < T\}},$$

and assume explicitly that this is a compact genus-zero surface, its lower boundary is one smooth Jordan curve, its upper boundary consists of m smooth root curves, all interior critical points are nondegenerate index-one saddles, the normalised gradient field

$$X = \frac{\nabla u}{|\nabla u|^2} = -\frac{p}{p'}$$

is transverse to the level boundaries, and no maximal X -trajectory has two saddle endpoints.

Problem 7.2 (finite strip decomposition after ray cuts). For every $\eta > 0$, construct disjoint Morse neighbourhoods N_j with $\sum_j \text{diam } N_j < \eta$ and a finite set of complete separatrix or regular-flow cuts so that every component of the complement is flow-diffeomorphic to a rectangle

$$[a_S, b_S] \times [0, 1], \quad u(\Phi_S(t, s)) = t,$$

with connected level sections and no uncut annular component. Give an explicit finite bound $E(s, m)$ for the number of strips, or exhibit the minimal missing hypothesis or a counterexample.

No particular formula such as $2s + 1$ is presumed. Boundary tangencies, simultaneous levels, branch reunion through an annulus and non-Hausdorff orbit spaces must be handled rather than suppressed.

3. Metric fan-in without losing the coefficient

For a strip S , write

$$\Gamma_t^S = S \cap \{u = t\}, \quad P_S(t) = \mathcal{H}^1(\Gamma_t^S),$$

and let k_S be its number of root ends. The flux identity

$$\int_{\Gamma_t^S} |\nabla u| ds = 2\pi k_S$$

gives the average trajectory estimate

$$\int_{\Gamma_{t_0}^S} \text{len}(\gamma_x) d\mu_{t_0}(x) = \frac{1}{2\pi k_S} \int_{a_S}^{b_S} P_S(t) dt.$$

Problem 7.3 (additive-error strip gluing). Assuming Problem 7.2, select trajectories in all strips and connect them through the saddle and root neighbourhoods so that, for every $\eta > 0$, the resulting embedded tree contains all m roots and obeys

$$\text{len}(G) \leq \frac{1}{2\pi} \int_{2\alpha}^{\infty} P_V(t) dt + \eta. \quad (6.1)$$

The total saddle, annular-cut and root-cap cost must be below η without a multiplicative loss in $1/(2\pi)$.

For the final strict inequality, use the actual collar slack

$$q = \frac{1}{2\pi} \int_{\alpha}^{2\alpha} P_V(t) dt > 0$$

and give budgets that keep the perturbation, tree error and transfer cost below fixed fractions of q . Independently choosing a shortest trajectory in each strip is not enough unless the attachment mismatch is controlled.

4. Coefficient perturbation and stability

The constant-translation stage is no longer open. Once a finite critical-value family is injective, Lean proves an arbitrarily small translation making every value nonzero and pairwise positive-ray separated (*exists small translation separating arguments*). It also proves the explicit root-retention estimate; a shift below ε keeps all roots in the unit disc when

$$((n+1)\varepsilon)^{1/n} + \rho < 1$$

(*constant perturbation roots in unit disk*). A constant translation cannot separate initially equal critical values.

Problem 7.4 (two-stage generic perturbation with slack). For fixed root discs, compact collar K , regular levels $\alpha/2, \alpha, 2\alpha, 5\alpha/2$ and collar slack $q > 0$, prove that an arbitrarily small

$$g_{\lambda, \beta}(z) = f(z) + \lambda z + \beta$$

can be chosen so that:

1. $f + \lambda z$ has simple relevant critical points and injective complex critical values;
2. the checked constant translation β makes those values nonzero and pairwise ray-separated;
3. no critical point enters the protected collar or truncation boundary;
4. the relevant component remains in K , contains exactly the corresponding perturbed roots and has slack $q_g > q/2$; and
5. root displacement and straight-line transfer back to the original roots consume less than a prescribed fraction of q/m .

If the one-coefficient perturbation λz cannot ensure all five properties, give the smallest additional lower-coefficient direction that can, or an explicit obstruction.

Finite planar avoidance and the subsequent constant translation must not be relisted as missing; coefficient genericity, component stability and slack stability are the open content.

5. The global Newton-flow claim ceiling

Problem 7.5 (relative global Newton-flow theorem). On a compact regular band

$$M_{a,b} = \{z : a \leq -\log |p(z)| \leq b\},$$

prove real-time existence and the identities

$$p(z(t)) = e^{-(t-t_0)} p(z(t_0)), \quad u(z(t)) = u(z(t_0)) + t - t_0$$

throughout every maximal orbit; classify both limiting endpoints among the regular boundary, root ends and finite saddle set; and exclude recurrent, accumulating and non-Hausdorff orbit phenomena. Under pairwise ray separation of the critical values, decide whether the flow is Morse–Smale relative to the boundary or whether the strongest valid conclusion is only absence of saddle-to-saddle connections.

The checked algebra proves the pointwise value equation and the endpoint-ray consumer. It does not supply global solution theory or an orbit-space graph. A positive stronger theorem must give the graph and a finite edge bound; a negative answer should exhibit the simplest ray-separated polynomial carrying the remaining pathology.

Erdős #1041 remains open. The source now publicly verifies the Newton kernel, finite ray avoidance and quantitative constant-translation root control; the coefficient perturbation, corrected planar decomposition and metric gluing remain the exact unresolved producers.

Statements and declarations

Lean does not check the exposition, citation choices, or interpretation. This manuscript cites Lean only for the formal statements and proofs that the pinned kernel accepts. The checked core is the Newton value equation, the exponential first integral, the

consumer form of ray separation, the finite planar-avoidance theorem, quantitative constant-translation root retention, and the ray-collision parameterisation. The decomposition and length statements of §7 are not proved. The diagnosis of Proposition 12 in §5 concerns its printed local saddle construction; it does not refute the proposition's statement. The search results of §6 are computations.

A Guide to the formal sources

The public `ErdosProblems.Erdos1041.NewtonFlowRaySeparation` module contains the checked source for this note. The search of §6 is `scripts/search_counterexample.py` in the source package. The declaration table below is pinned to the shared formal-source commit used throughout this problem-note series.

- `newton flow vector`
- `derivative mul newton flow vector`
- `newton flow value has deriv at`
- `newton flow scaled value has deriv at zero`
- `same positive ray`
- `same positive ray refl`
- `same positive ray symm`
- `same positive ray trans`
- `translated same positive ray parameterization`
- `real affine line`
- `real affine line eq affine span`
- `is closed real affine line`
- `dense compl real affine line`
- `exists small avoiding finite real affine lines`
- `same positive ray imp mem real affine line`
- `exists small translation separating arguments`
- `norm lt one of near root`
- `constant perturbation root near original`
- `constant perturbation roots in unit disk`
- `same positive ray of real exp decay`
- `no newton connection of not same positive ray`

References

- [1] T. F. Bloom, *Erdős Problems*, problem 1041. <https://www.erdosproblems.com/1041>
- [2] P. Erdős, F. Herzog, and G. Piranian, *Metric properties of polynomials*, J. Analyse Math. **6** (1958), 125–148. <https://doi.org/10.1007/BF02790232>
- [3] S. Ghosh and K. Ramachandran, *Number of Components of Polynomial Lemniscates: A Problem of Erdős, Herzog, and Piranian*, arXiv:2312.13673v1 (2023). <https://arxiv.org/abs/2312.13673>
- [4] S. Sutherland, *Bad Polynomials for Newton’s Method*, in B. Bielefeld and M. Lyubich (eds.), *Conformal Dynamics Problem List*, Stony Brook IMS preprint (1992), 42–44. <https://www.math.stonybrook.edu/preprints/ims92-7.pdf>
- [5] shtuka, *A Short Path Joining Two Zeros Inside a Polynomial Lemniscate*, manuscript posted 24 March 2026, 48 pp. <https://shtuka123.github.io/1041/main.pdf>. The file at this URL has since been replaced by a shorter partial version that no longer contains Proposition 12; the durable public record of the March version, its defect, and the author’s 26 March 2026 concession is the discussion thread at <https://www.erdosproblems.com/forum/thread/1041>.
- [6] V. S. Pendyala, *A Degree-Four Lemniscate Path Theorem*, arXiv:2606.24875v1 (2026). <https://arxiv.org/abs/2606.24875>, doi:10.48550/arXiv.2606.24875.

Statement	Status	Exact boundary
Erdős #1041	Open	No proof is claimed.
Newton value equation $w' = -w$	Checked	Away from critical points, along any trajectory tangent to $-f/f'$.
Exponential first integral	Checked	$\frac{d}{dt}(e^t f(z(t))) = 0$.
Ray separation of critical values	Checked (consumer form)	Endpoints of a finite connection share one oriented ray; distinct rays exclude a connection.
Ray-collision locus	Checked	$\beta = (ra - b)/(1 - r)$, $r > 0$, $r \neq 1$: one real parameter per pair.
Quartic case	Cited	Proved in [6, Thm. 1, p. 1]; does not extend to general degree.
Translated quartic quotient fibres	Ordinary theorem with a Lean-checked metric kernel	For $f(z) = P((z - h)^q)$, P monic quartic and $q \geq 2$; Pendyala supplies the quartic geometry, while Lean checks the root-lift density, exact primitive, and strict endpoint budget.
Signed-moment cyclic tetranomials	Lean-checked two-index safe-spoke theorem	For an indexed finite root family of $g(w) = w^m + aw^r + bw^s + c$, an exact signed L^2 moment budget selects two distinct indices whose complete spokes are safe; distinct root values require injectivity of the indexing map.
Concyclic zeros with $2\rho^n \leq 1$	Ordinary proof; finite exact and numerical checks	Distinct-root theorem; not Lean checked; the unrestricted concyclic case remains open.
Unrestricted proof of [5, Theorem 1, p. 1]	Proof gap	Proposition 12 uses a false three-ended local saddle block; located publicly by Tao (25 March 2026), conceded by the author at statement level (26 March 2026). No counterexample is exhibited.
Constant-translation ray separation and root retention	Checked	After critical-value injectivity, arbitrary small ray avoidance and an explicit unit-disc margin.
Coefficient perturbation and slack stability	Open	Must first create injective critical values and preserve the component, collars and length budget.
Reeb decomposition and length fan-in	Open	The two surviving producers.
Random search to degree 10	Verified finite instances	Upper bounds for sampled configurations only.